

Physical Chemistry (I)

Lecture 13



In the Last Class

- Chemical potentials
 - Chemical potentials
 - Meanings
- Thermodynamics potentials
 - Differential forms
 - Properties of exact differentials
- Thermochemistry of Gibbs energy



Phase Transition

Spontaneous conversion
of one phase into another phase

- occurs at transition temperature T_{trs} (for given p)
- At T_{trs} , the two phases are at equilibrium

Review

Equilibria and Chemical Potential

$$dU = TdS - pdV + \mu dn$$

- Mechanistic equilibrium: $p_1 = p_2$
 - Conjugate extensive variable: dV
- Thermal equilibrium: $T_1 = T_2$
 - Conjugate extensive variable: $dS = \delta q/T$
- Diffusive equilibrium: $\mu_1 = \mu_2$
 - Conjugate extensive variable: dn



μ as Molar Gibbs Energy

$$\mu = \left(\frac{\partial G}{\partial n} \right)_{T,p} = \frac{G}{n}$$

$$G = \mu n$$

$$A = \mu n + TS$$

$$H = \mu n - pV$$

$$U = \mu n + TS - pV$$

Gibbs-Duhem Equation

$$U = \mu n + TS - pV$$

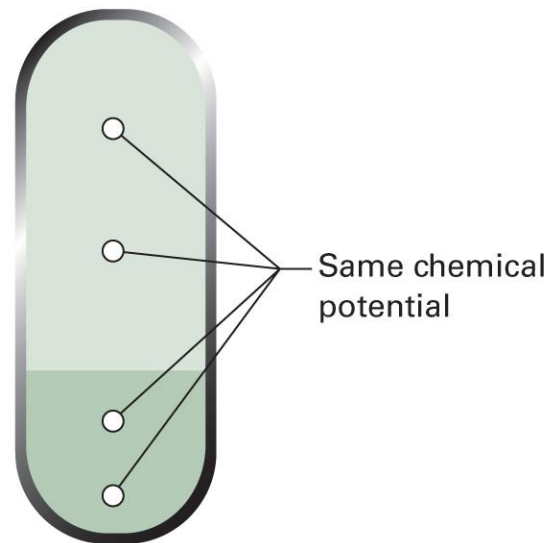
$$SdT - Vdp + nd\mu = 0$$

(Gibbs-Duhem equation)

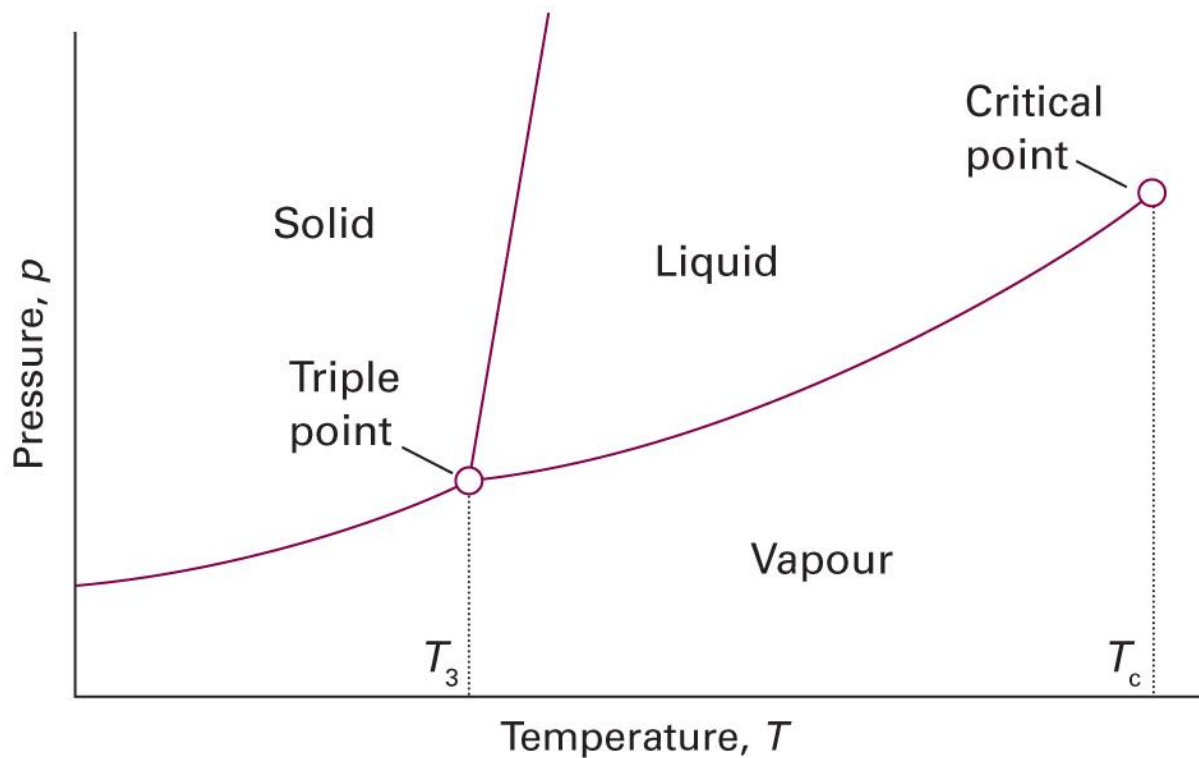
$$d\mu = -S_m dT + V_m dp$$

Criteria for Phase Stability

At equilibrium, the **chemical potential** of a substance is the **same** in and throughout every phase in the system



Phase Diagram



Phase Rule

“Can we have a quadruple point?”

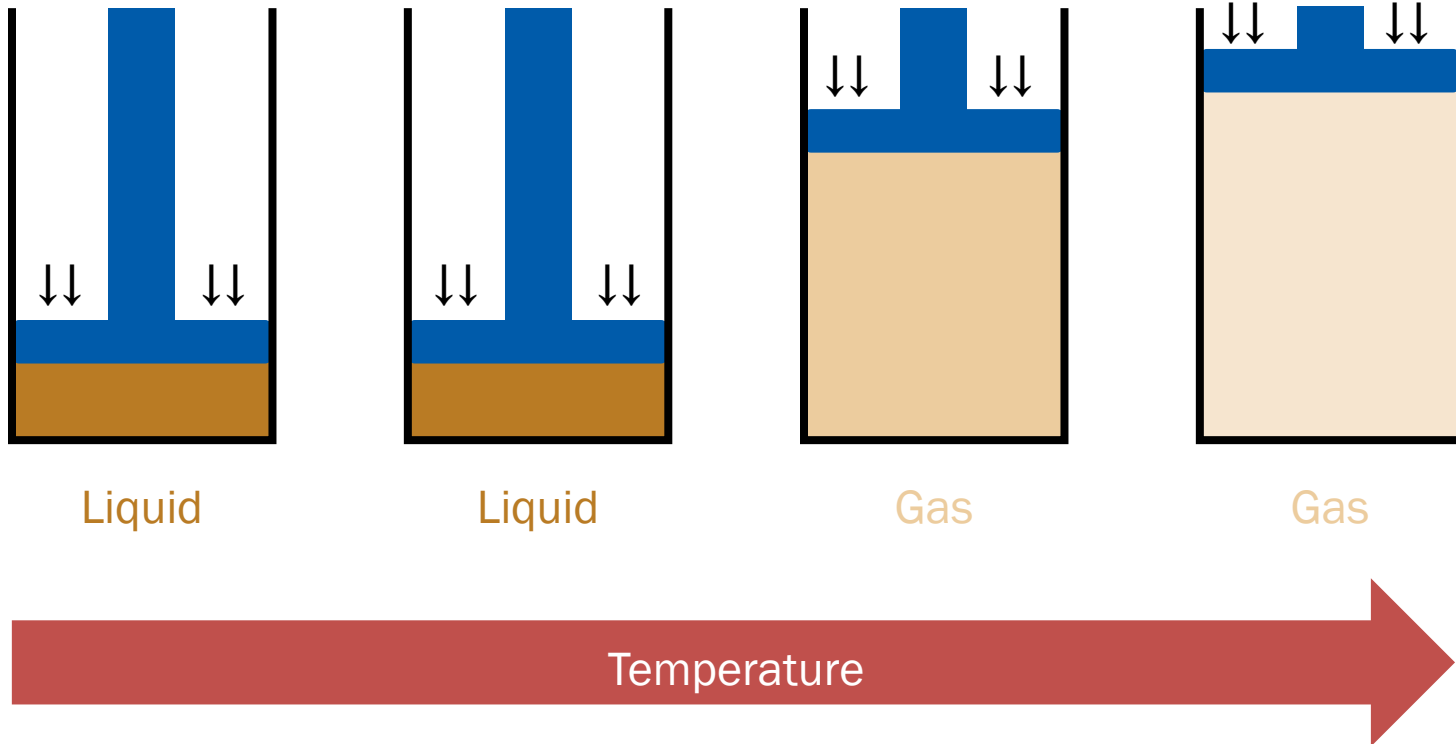
- **Phase rule:** a general relation between
 - Number of degrees of freedom F
 - Number of components C
 - Number of phases at equilibrium P

$$F = C - P + 2$$

Not in Atkins

Liquid-Gas Transition 1

(Isobaric change)

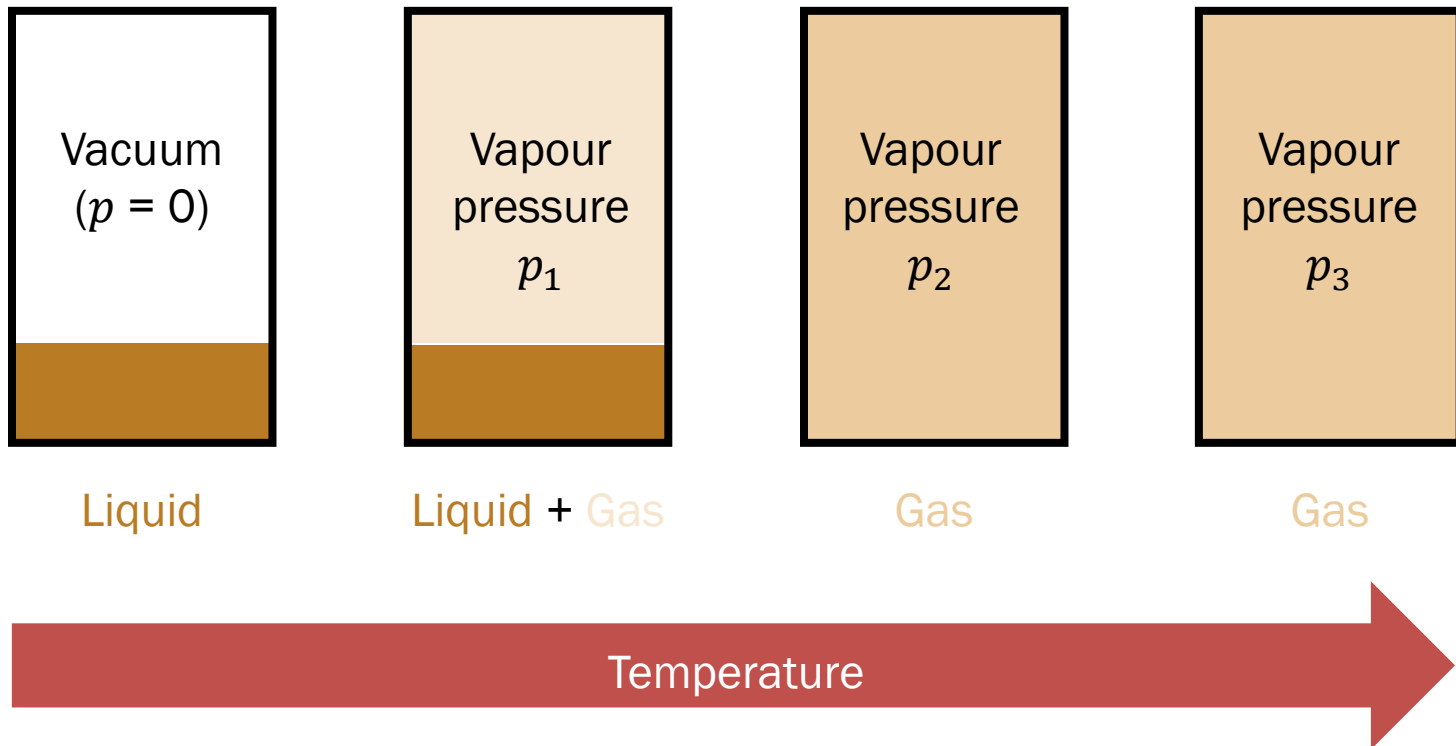


Not in Atkins

Liquid-Gas Transition 2

(Isochoric change)

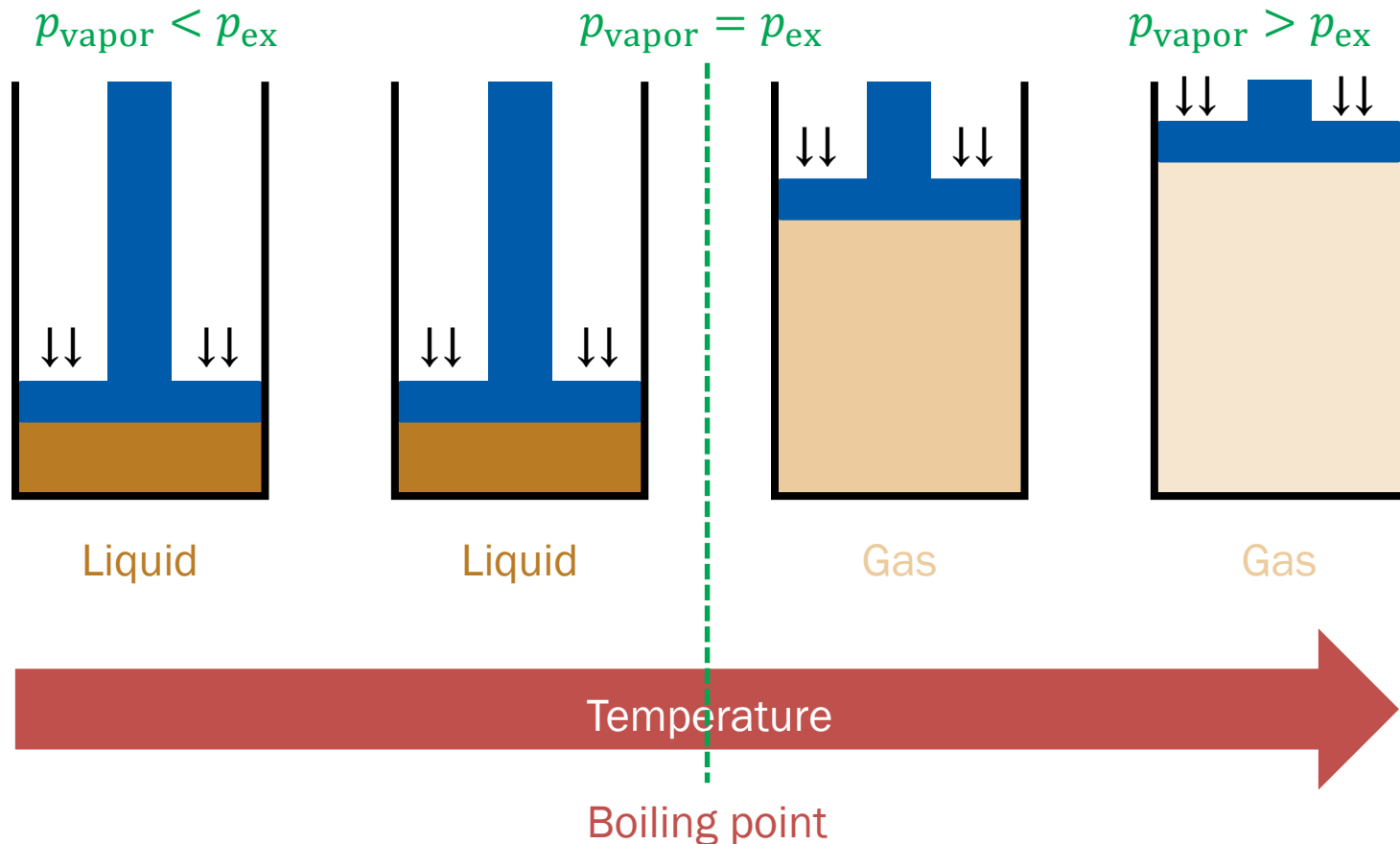
$(p_1 < p_2 < p_3)$



Not in Atkins

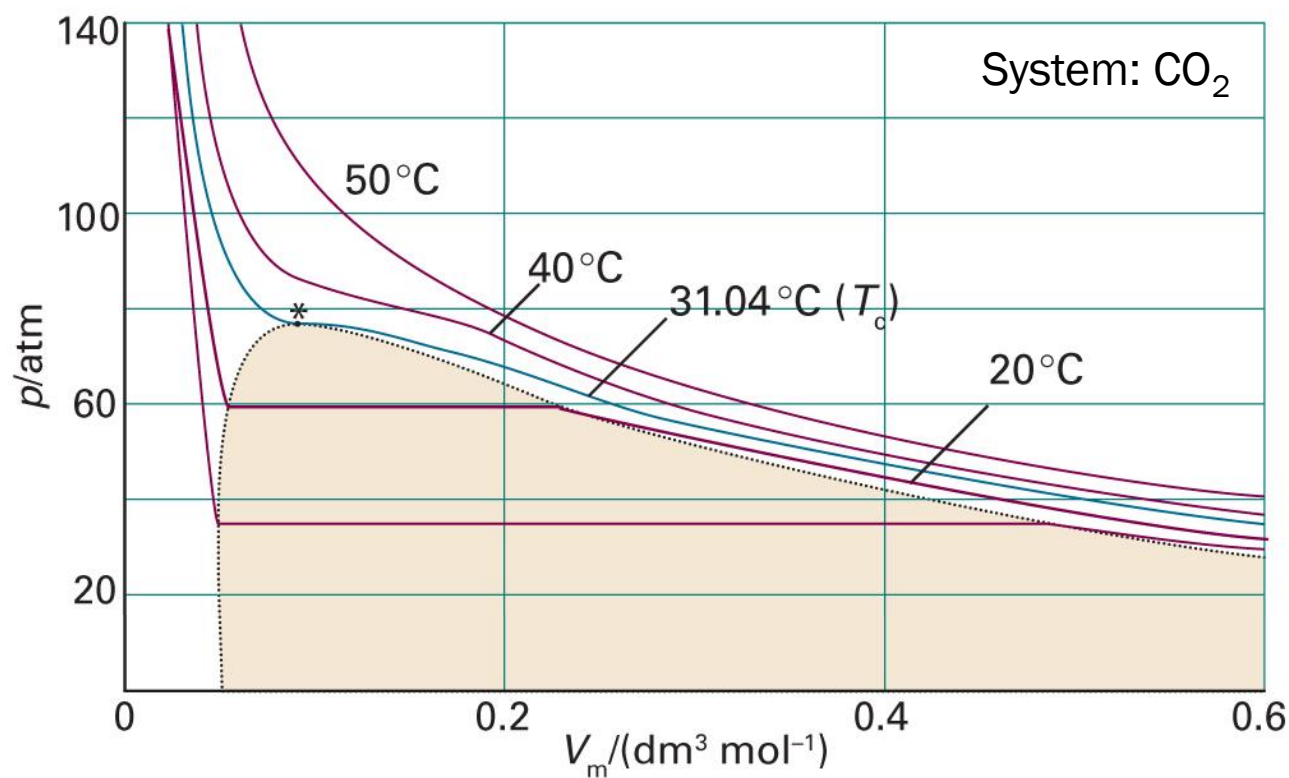
Vapor Pressure and Boiling

(Isobaric change)



Liquid-Gas Transition 3

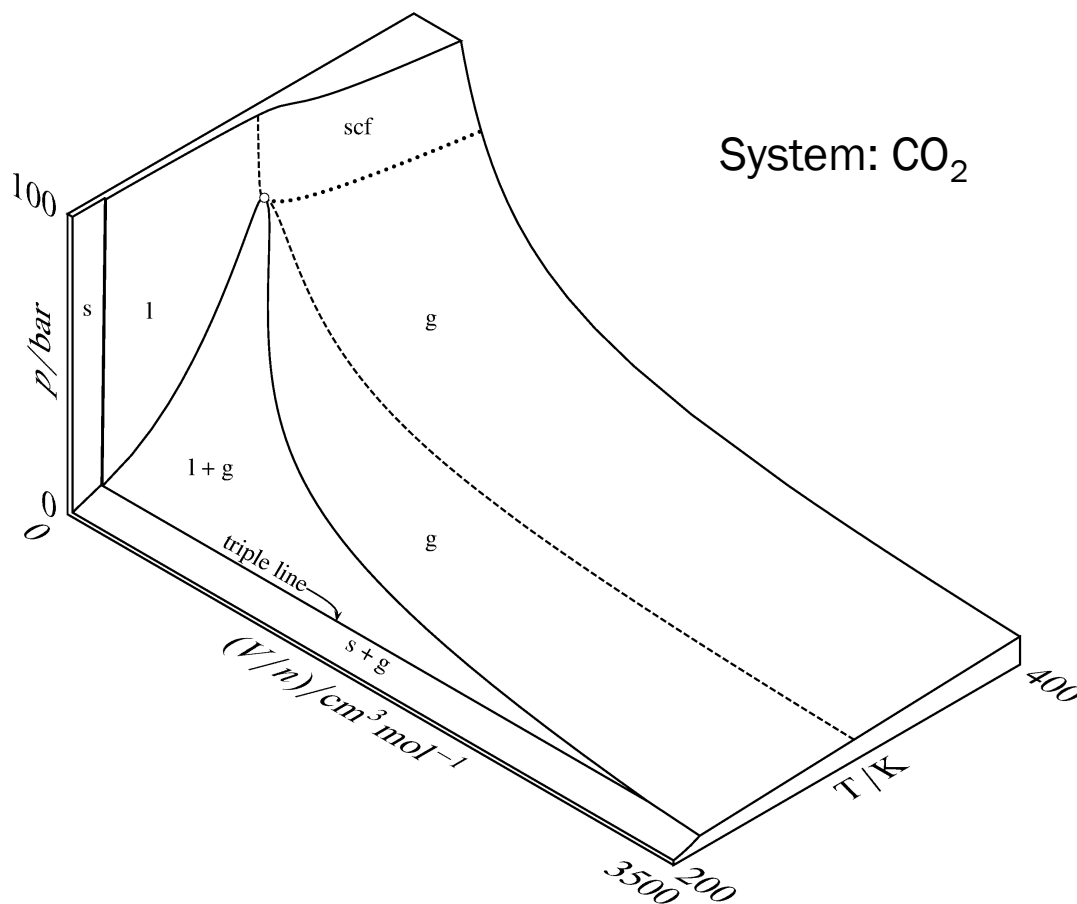
(Isothermal change)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 1C.2)

Not in Atkins

3-Dimensional Surface



(Image: <https://chem.libretexts.org/@go/page/20608>)

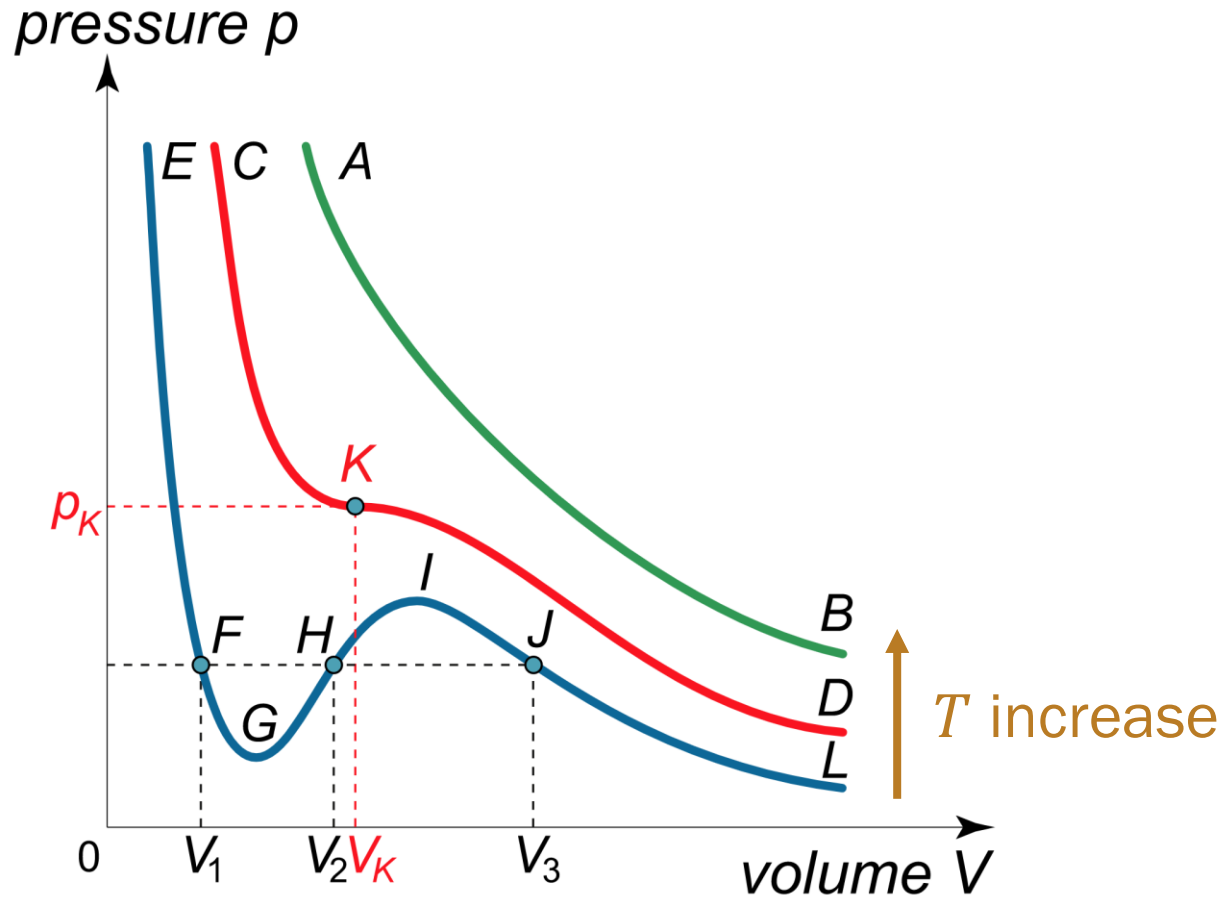
Good Old Friend

“Can we describe the liquid-gas transition using a model system?”

The van der Waals fluid!

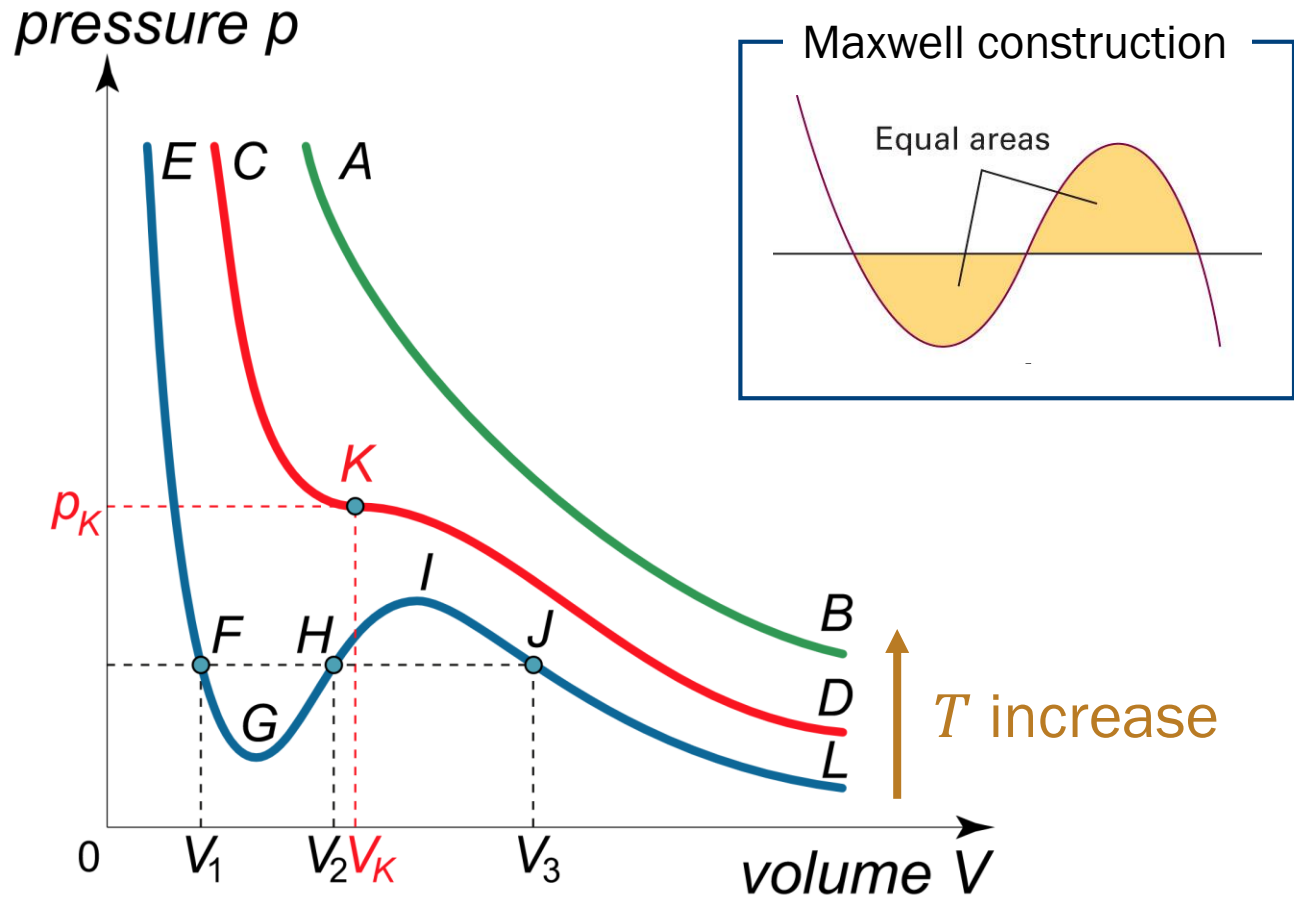
$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

van der Waals Isotherms



(Image: <https://www.math24.net/wp-content/uploads/2018/11/isotherms-of-van-der-waals-equation.svg>)

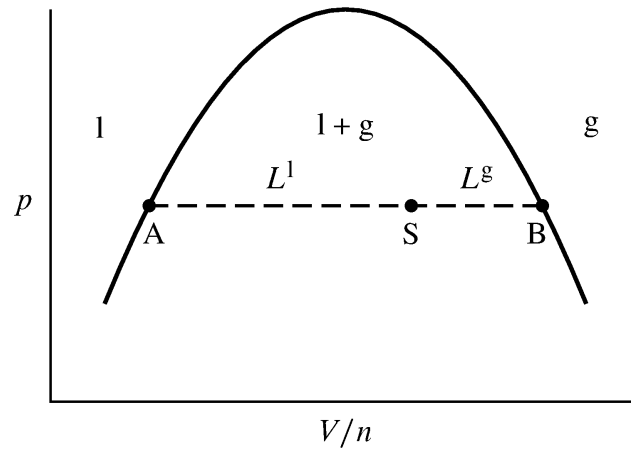
Curve E-L: Liquid-Gas Transition



(Image: <https://www.math24.net/wp-content/uploads/2018/11/isotherms-of-van-der-waals-equation.svg>)

Not in Atkins

What Happens on Tie Line?



- Total amount at point S: $n = n_l + n_g$
- Total volume at point S: $V = V_{m,l}n_l + V_{m,g}n_g$

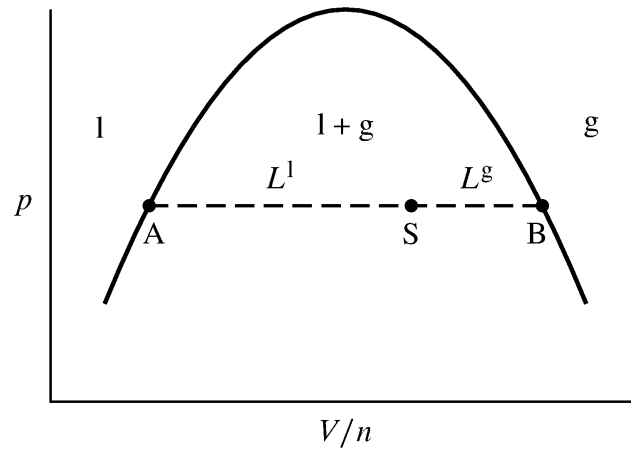
$$\frac{V}{n} = \frac{V_{m,l}n_l + V_{m,g}n_g}{n_l + n_g}$$



(Image: <https://chem.libretexts.org/@go/page/20608>)

Not in Atkins

What Happens on Tie Line?



$$n_l \left(V_{m,l} - \frac{V}{n} \right) = n_g \left(\frac{V}{n} - V_{m,g} \right)$$

$$n_l L^l = n_g L^g$$

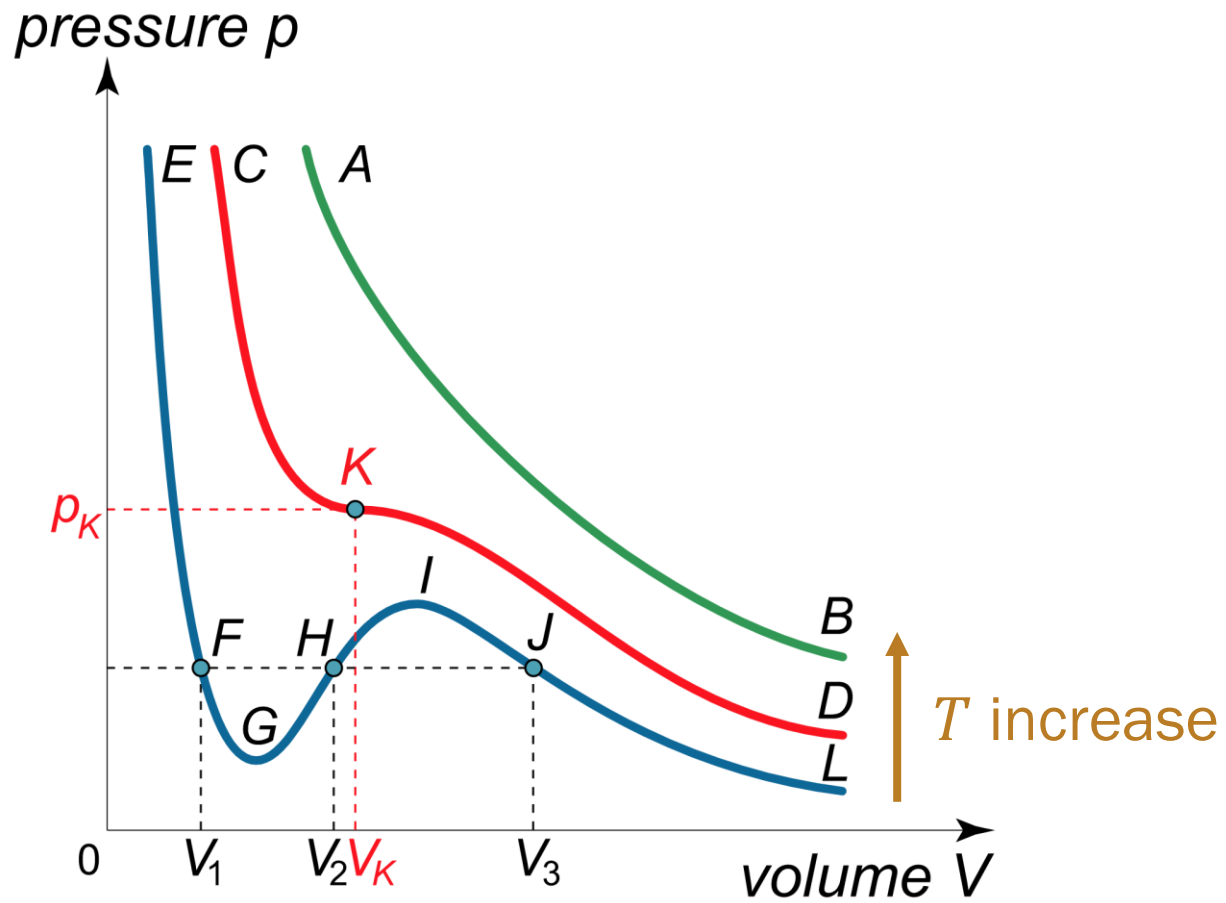
$$n_l : n_g = L^g : L^l \text{ (lever rule)}$$



(Image: <https://chem.libretexts.org/@go/page/20608>)

Not in Atkins

F-G and I-J: What Happens?



(Image: <https://www.math24.net/wp-content/uploads/2018/11/isotherms-of-van-der-waals-equation.svg>)

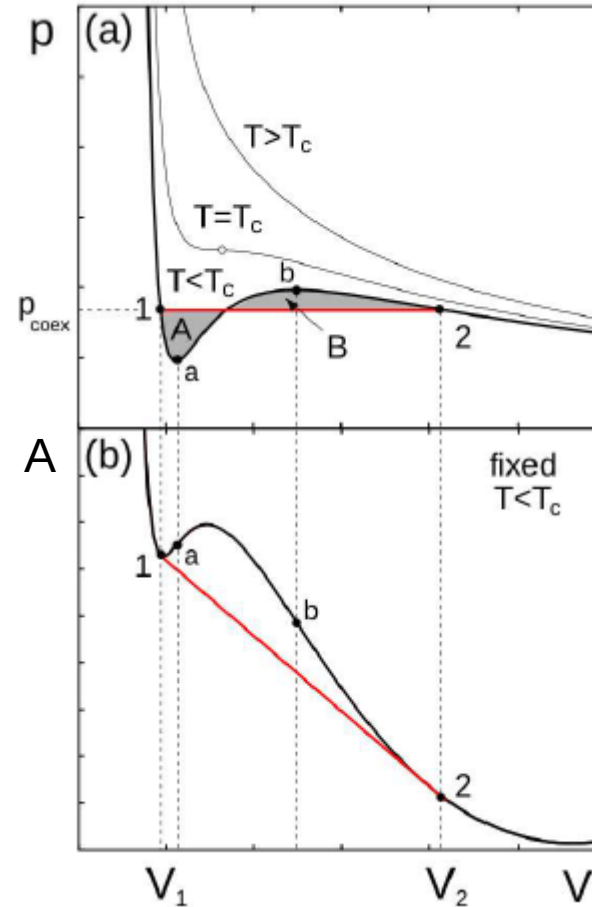
Not in Atkins

Metastable Phases

$$p = - \left(\frac{\partial A}{\partial V} \right)_{T,n}$$

At constant T and n ,

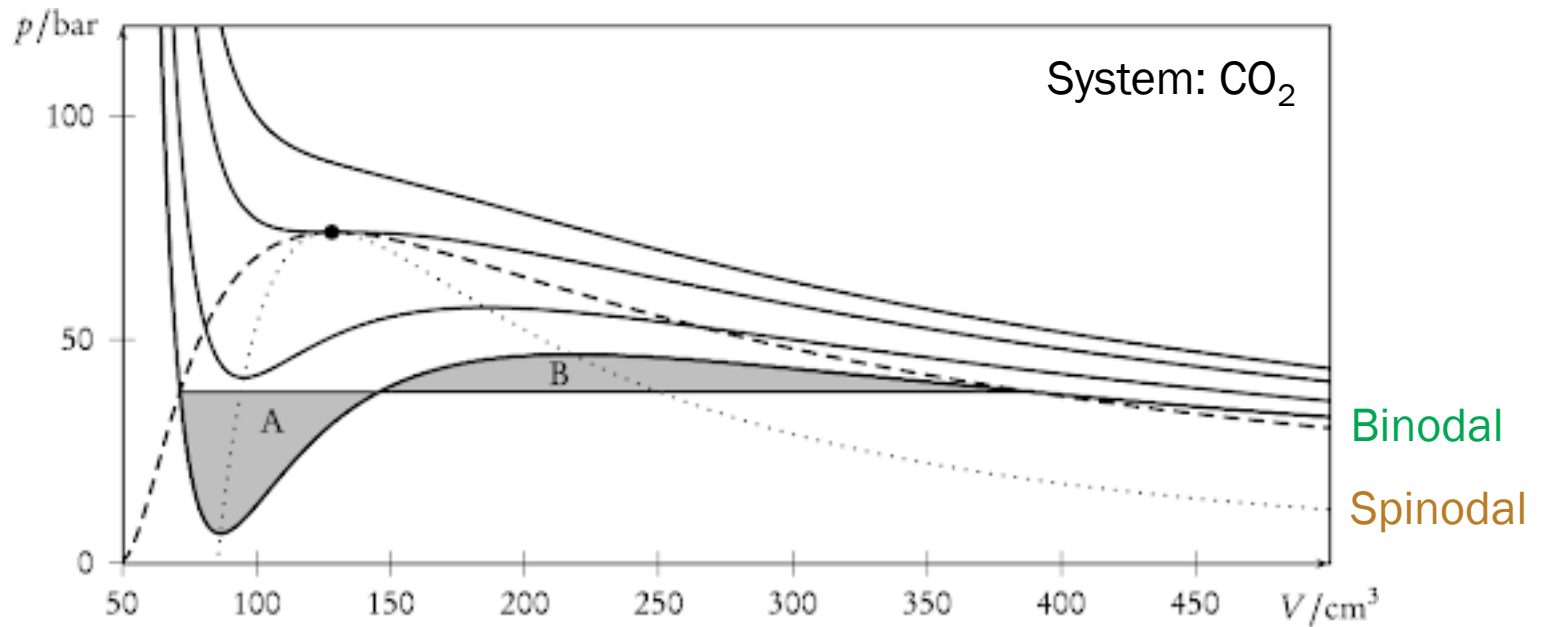
$$A = - \int p dV$$



(Image: <https://www.physicsforums.com/threads/liquid-gas-phase-transition-metastable-mixed-states.935905>) 21

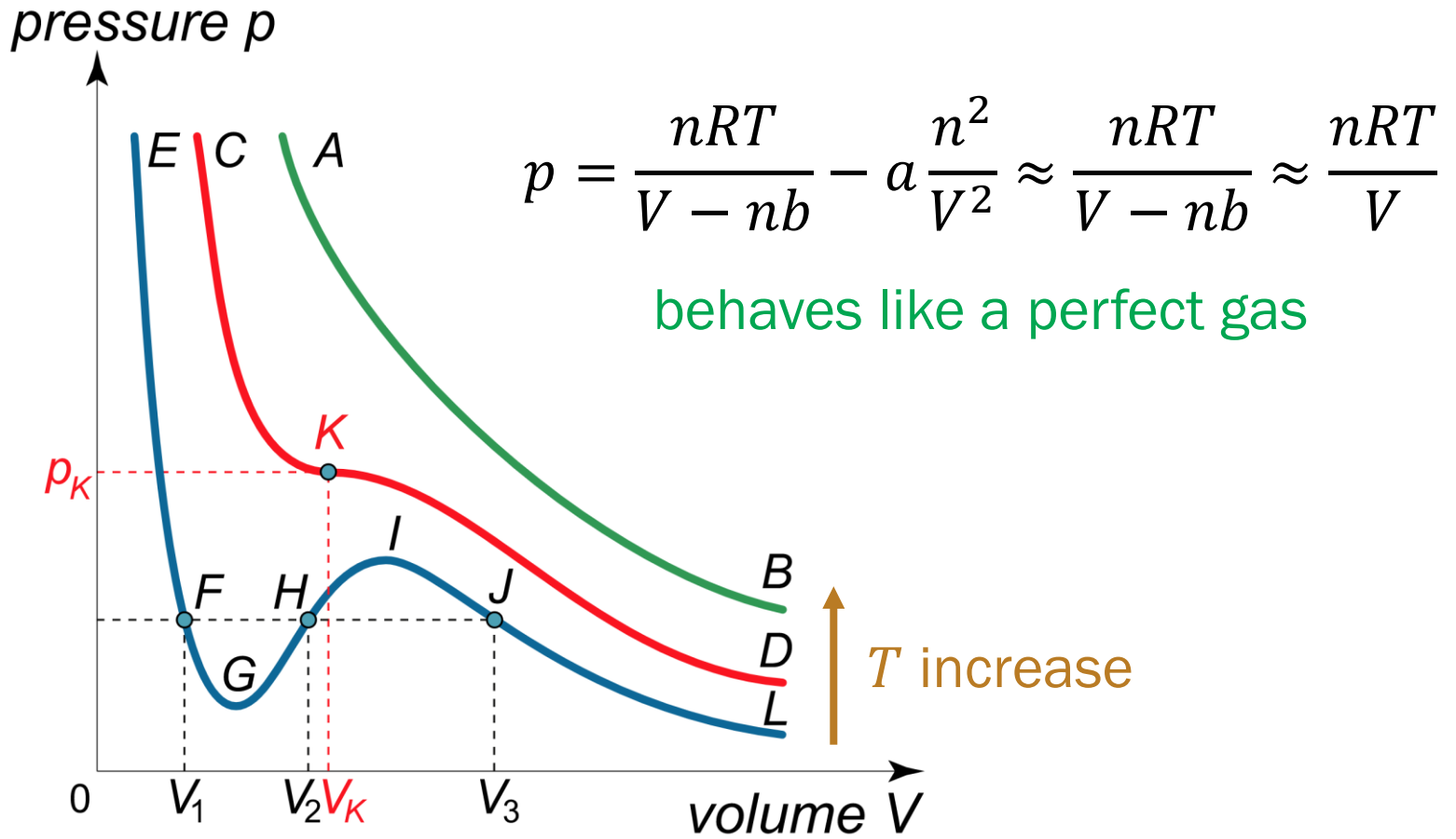
Not in Atkins

Binodal and Spinodal Lines



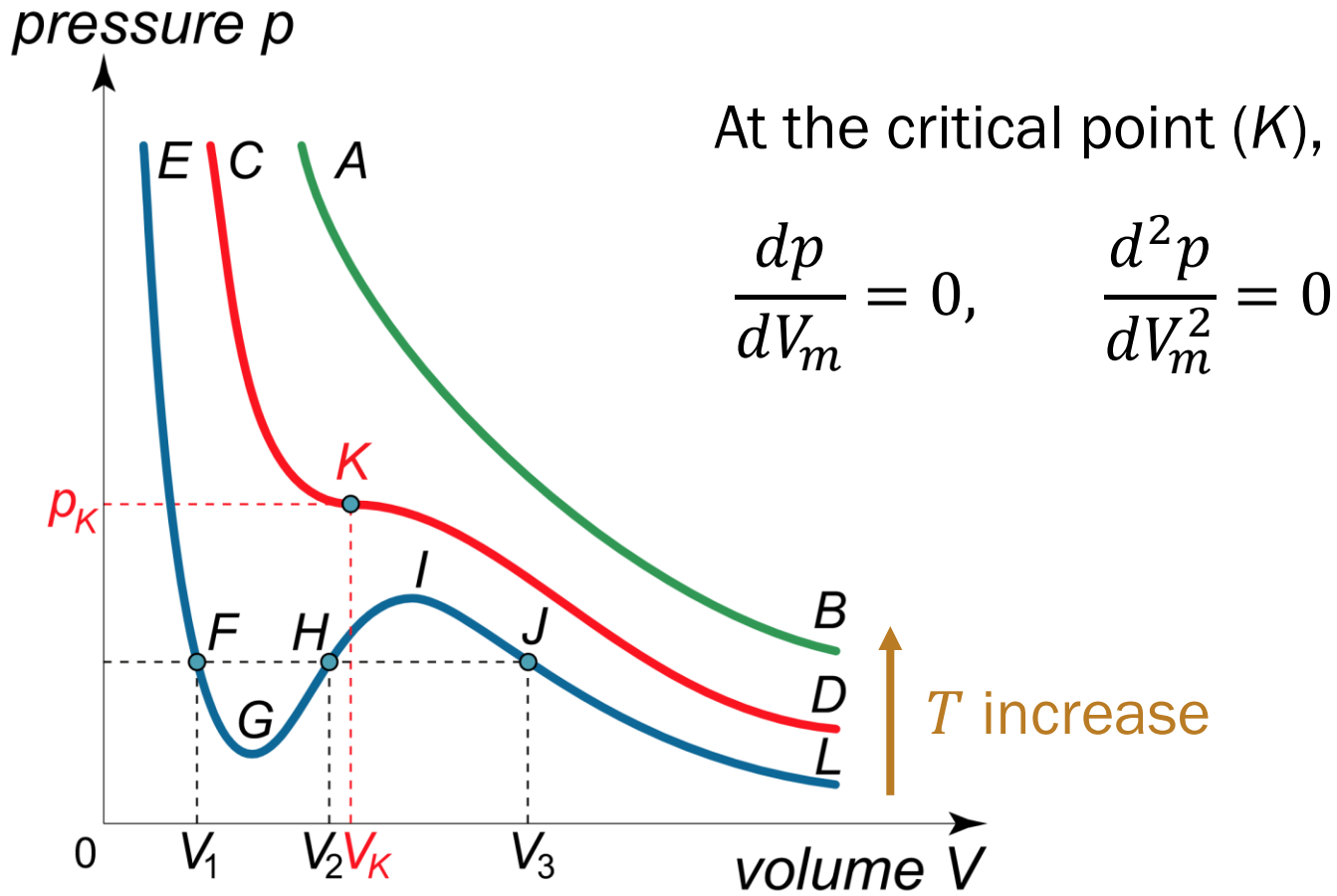
(Image: <http://www.mhkoeopf.de/Notes/Spinodal/spinodal.html>)

Curve A-B: Supercritical Fluid



(Image: <https://www.math24.net/wp-content/uploads/2018/11/isotherms-of-van-der-waals-equation.svg>)

Curve C-D: Critical Point



Critical Constants

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2} = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$\frac{dp}{dV_m} = -\frac{RT}{(V_m - b)^2} + \frac{2a}{V_m^3} = 0$$

$$\frac{d^2p}{dV_m^2} = \frac{2RT}{(V_m - b)^3} - \frac{6a}{V_m^4} = 0$$

Solving these equations, one can obtain

$$V_c = 3b, \quad p_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27bR}$$

Critical Compression Factor

$$V_c = 3b, \quad p_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27bR}$$

The critical compression factor is defined as

$$Z_c = \frac{p_c V_c}{RT_c} = \frac{3}{8} = 0.375$$

Experimental values of Z_c :

Ar – 0.292, CO₂ – 0.274, He – 0.305, O₂ – 0.308

Reduced Variables

$$p_r = p/p_c$$

$$V_r = V_m/V_c$$

$$T_r = T/T_c$$

Variables without the substance-specific features

Rewriting vdW Equation

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

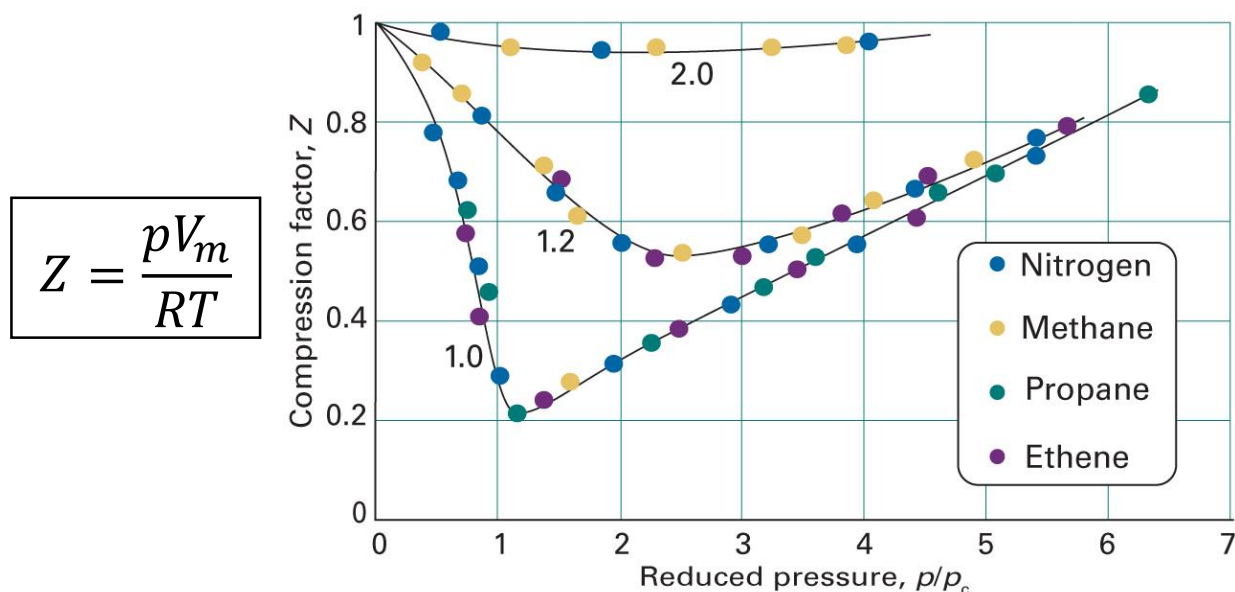
$$p_r p_c = \frac{RT_r T_c}{V_r V_c - b} - \frac{a}{V_r^2 V_c^2}$$

$$\frac{ap_r}{27b^2} = \frac{8aT_r}{27b(3bV_r - b)} - \frac{a}{9b^2V_r^2}$$

$$p_r = \frac{8T_r}{3V_r - 1} - \frac{3}{V_r^2}$$

Principle of Corresponding States

Gases at the same V_r and T_r exert the same p_r



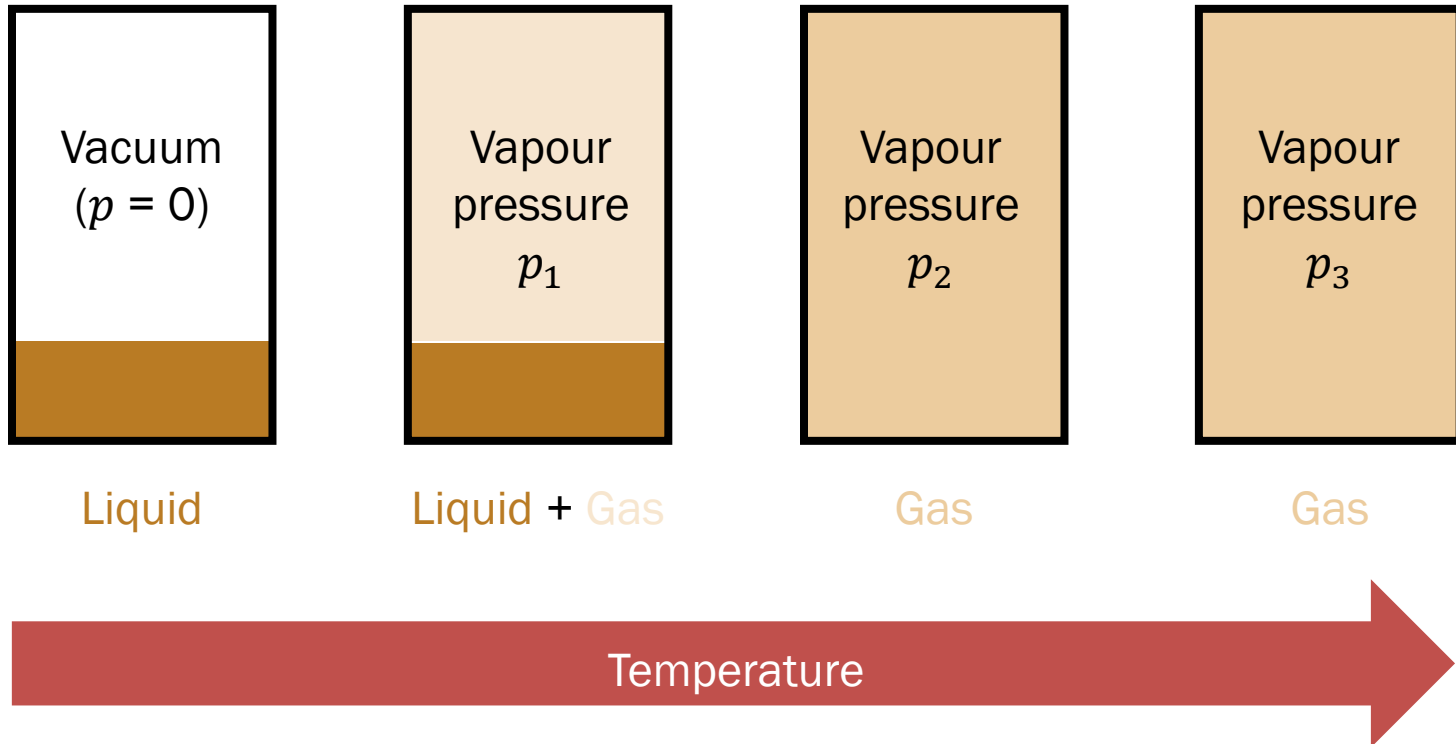
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 1C.8)

Not in Atkins

Coexistence Region

(Isochoric change)

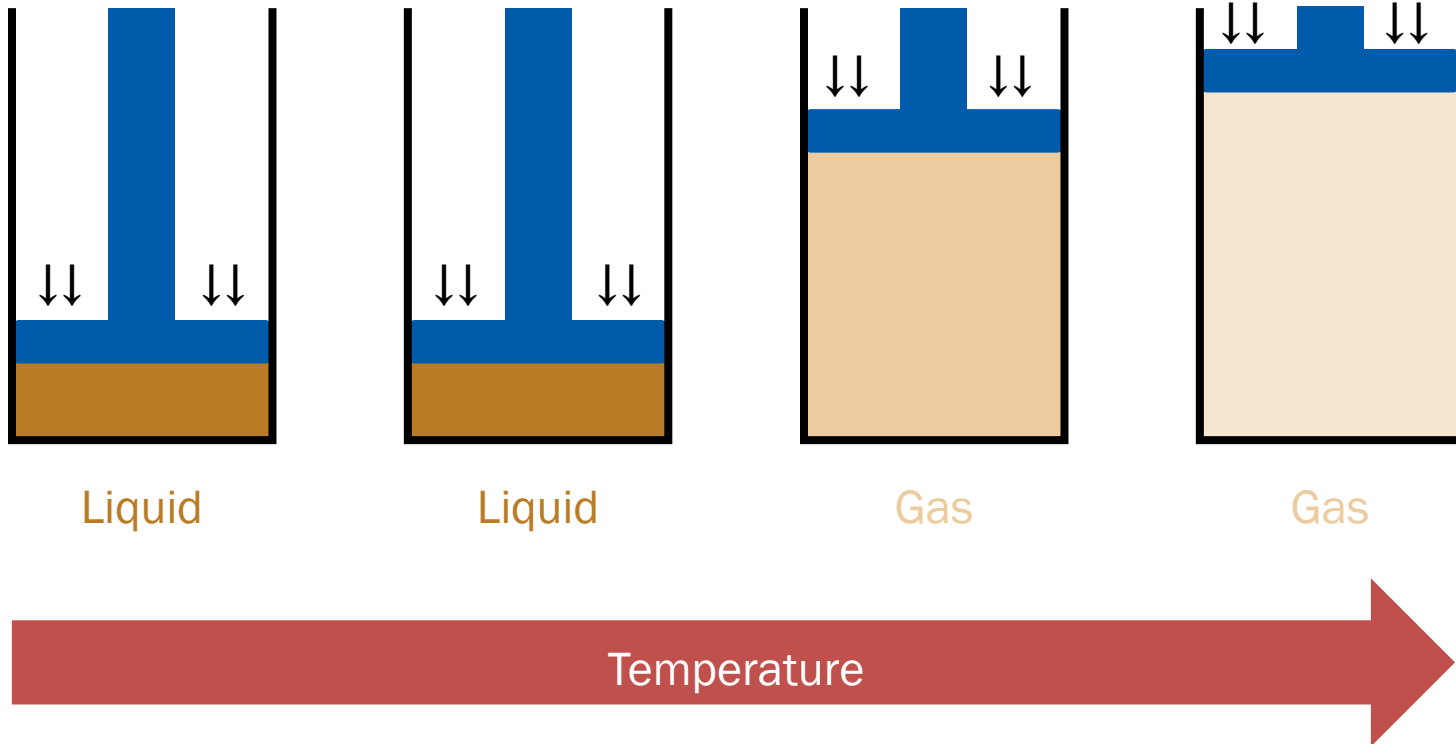
$(p_1 < p_2 < p_3)$



Not in Atkins

No Coexistence Region

(Isobaric change)



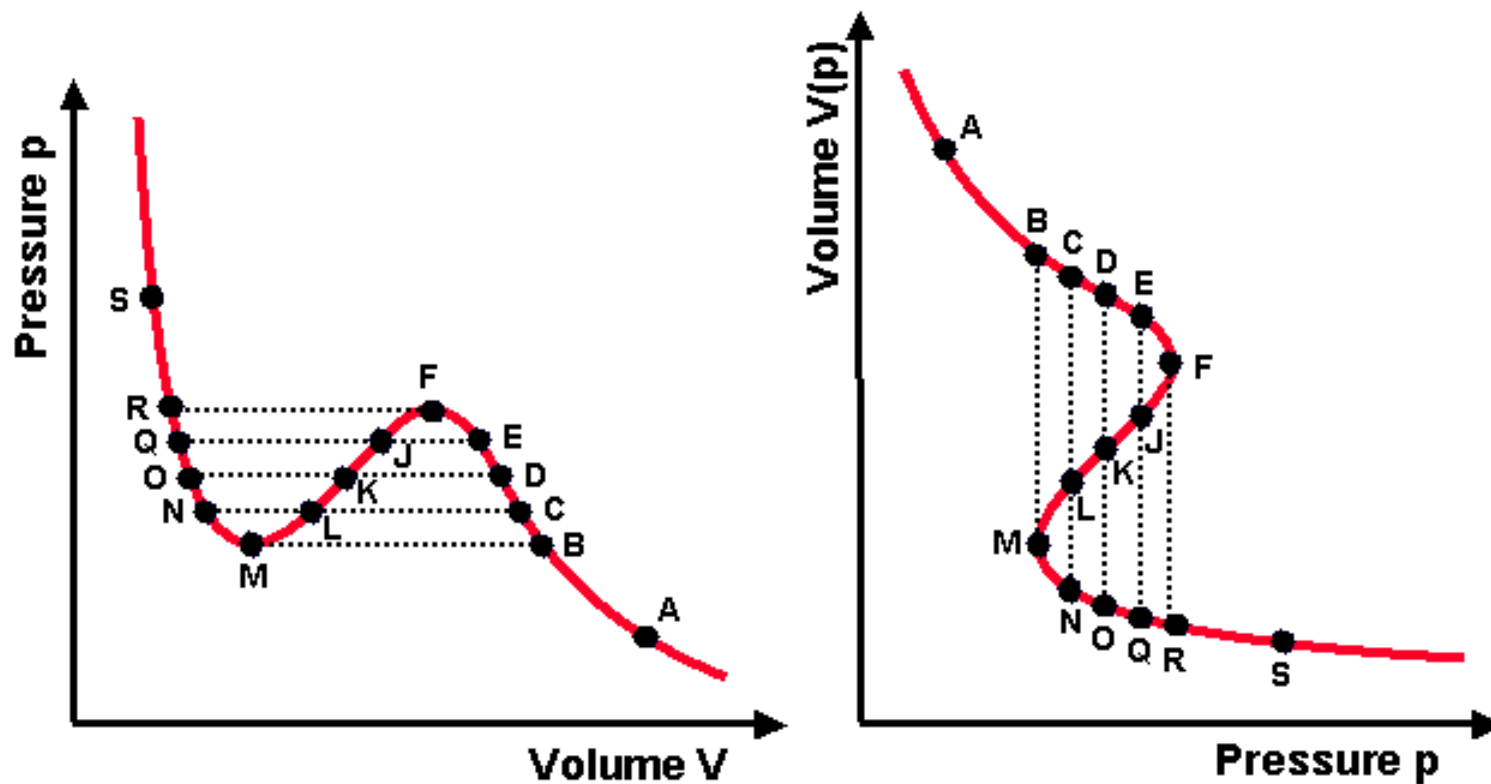
Not in Atkins

Transitions of vdW Fluid

Fixed	Independent	Dependent	Coexistence?
V	T	p	Yes
p	T	V	No
T	V	p	Yes
T	p	V	?

Not in Atkins

From $p(V)$ to $V(p)$



(Image: http://casey.brown.edu/chemistry/misspelled-research/crp/Edu/Documents/00_Chem201/7_phase_equil_1/Chem201_7_phase_equil_1.htm)

Not in Atkins

Gibbs Energy of vdW Fluid

$$V = \left(\frac{\partial G}{\partial p} \right)_{T,n}$$

At constant T and n ,

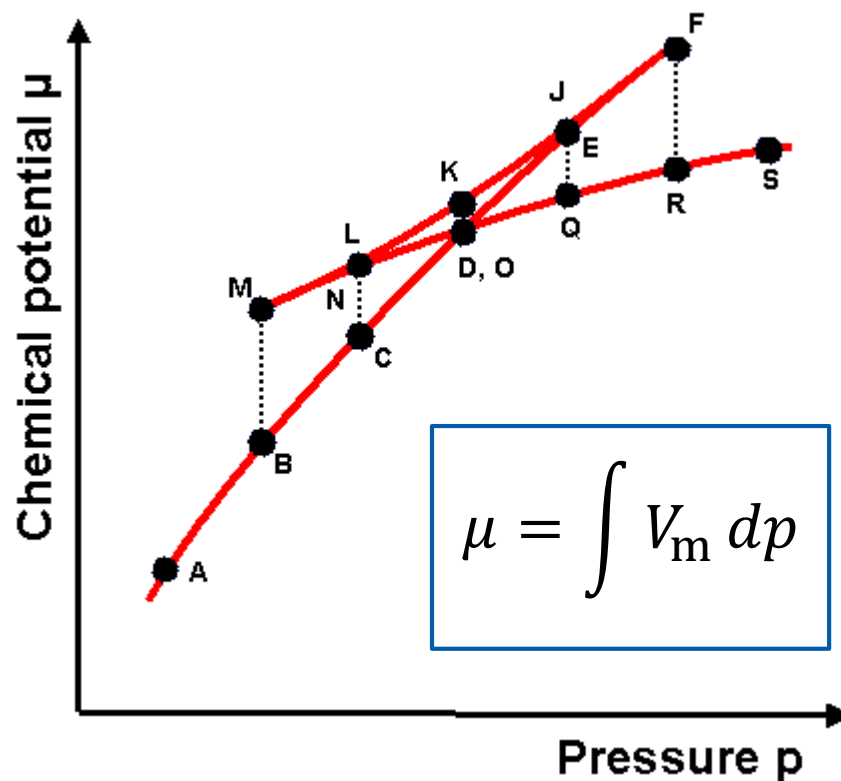
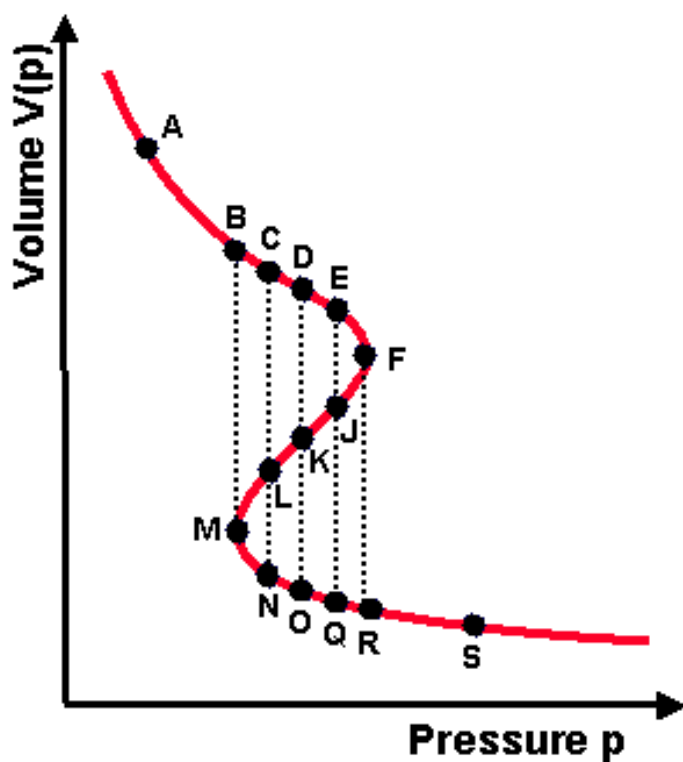
$$G = \int V dp$$

or,

$$\mu = \int V_m dp$$

Not in Atkins

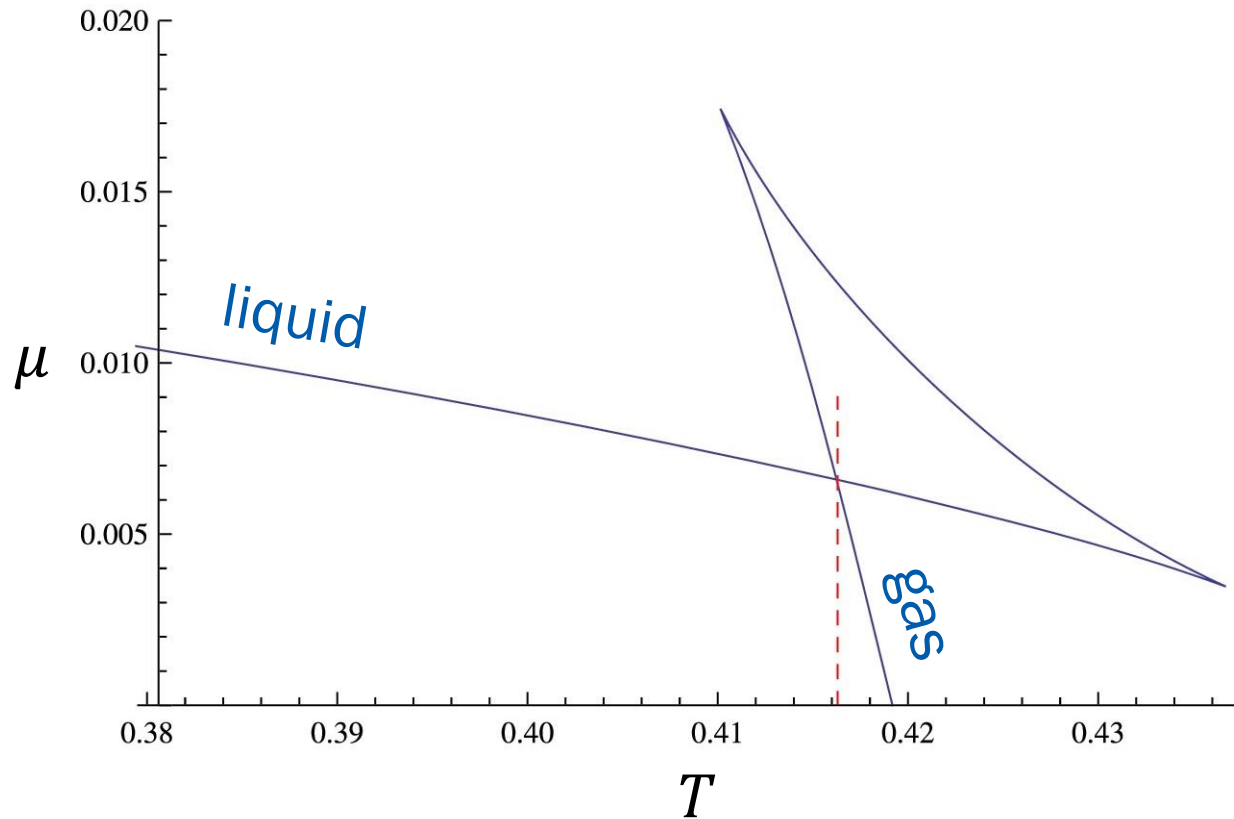
Chemical Potential $\mu(p)$



$$\mu = \int V_m dp$$

Not in Atkins

$\mu(T)$ of vdW Fluid



(Image: He et al., *Nuclear Physics B* 915: 243-261 (2017), Figure 3)

Generalization

Use **thermodynamics** of chemical potentials to study general properties of phase transitions

1. Phase transitions on (T, p) -space
2. Effect of external pressure on vapor pressure
3. Location of phase boundaries

Use of Gibbs-Duhem Equation

Recall

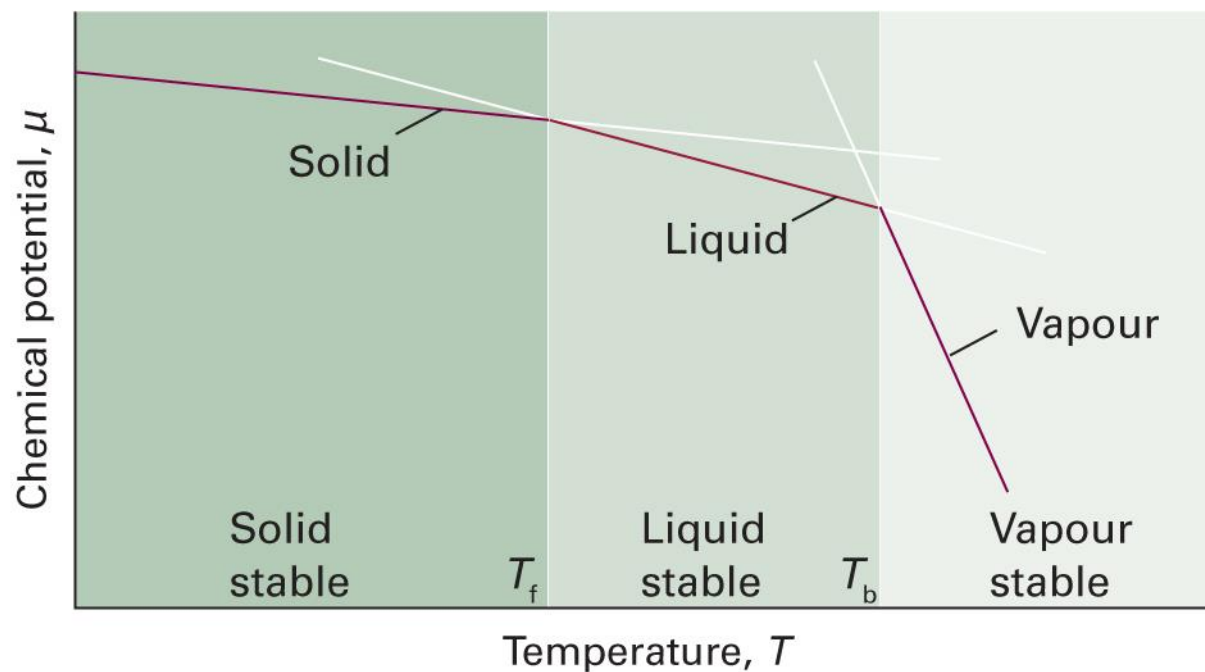
$$d\mu = -S_m dT + V_m dp$$

which gives

$$\left(\frac{\partial \mu}{\partial T}\right)_{p,n} = -S_m, \quad \left(\frac{\partial \mu}{\partial p}\right)_{T,n} = V_m$$

Chemical Potential and T

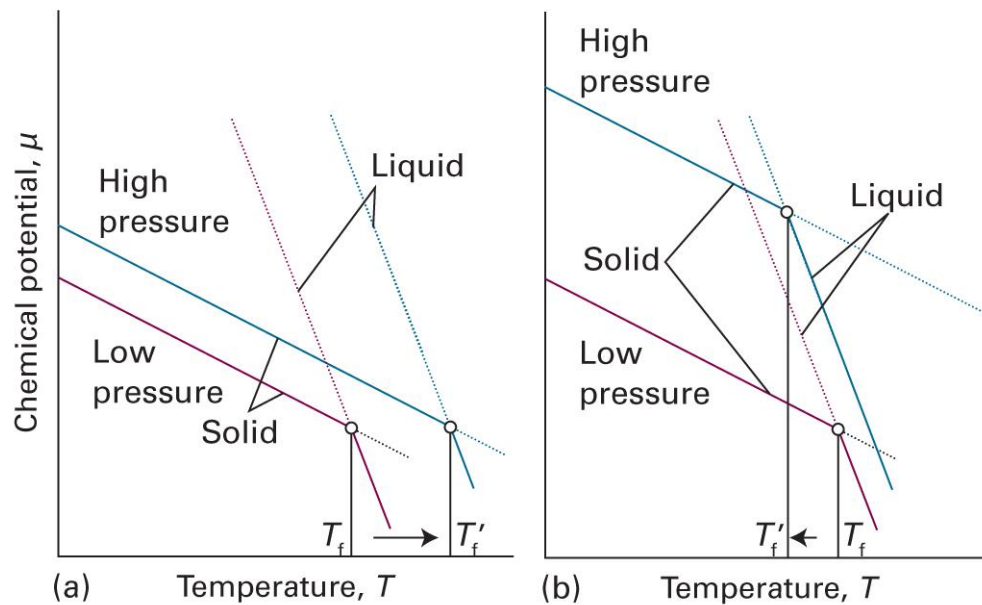
$$\left(\frac{\partial \mu}{\partial T}\right)_{p,n} = -S_m$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.1)

Chemical Potential and p

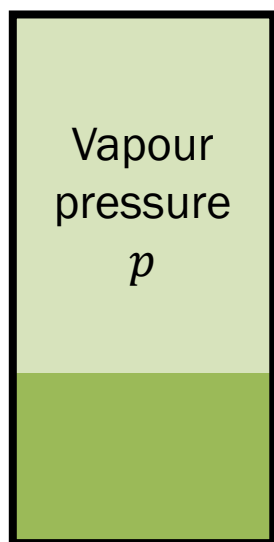
$$\left(\frac{\partial \mu}{\partial p}\right)_{T,n} = V_m$$



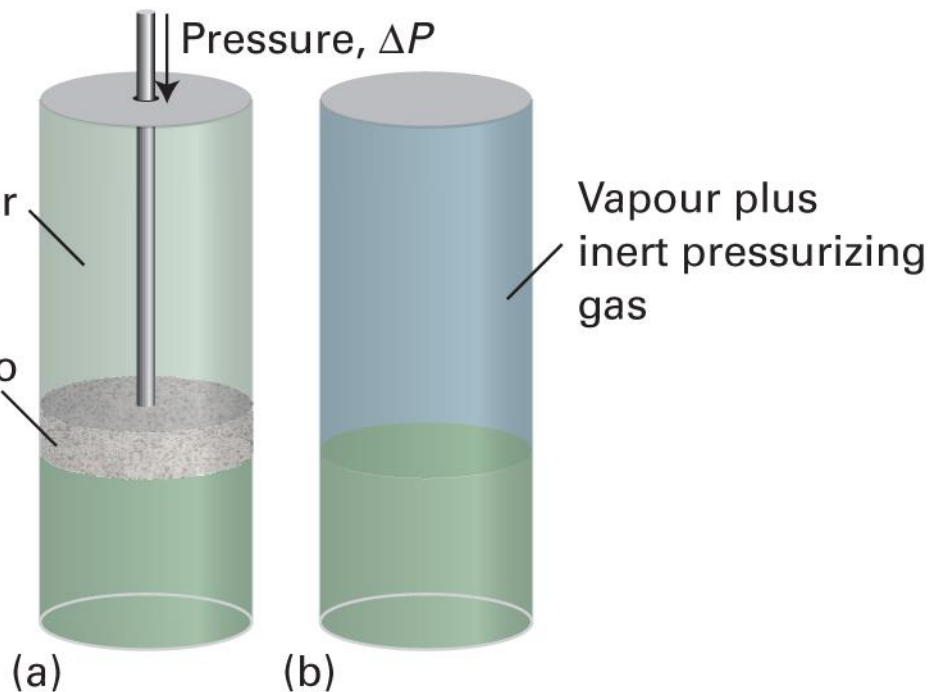
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4B.2)

Pressure on Liquid-Gas Eq.

Liquid + Gas



Piston
permeable to
vapour but
not liquid



Liquid-Gas Equilibrium

At equilibrium,

$$\mu(l) = \mu(g)$$

$$d\mu(l) = d\mu(g)$$

Since $d\mu = V_m dp$,

- Liquid: $d\mu(l) = V_m(l)dP$ (P : external pressure)
- Gas: $d\mu(g) = V_m(g)dp = (RT/p)dp$ (p : vapor pressure)

$$\frac{RT}{p} dp = V_m(l)dP$$

Integration

$$\frac{RT}{p} dp = V_m(l) dP$$

$$\int_{\text{state } i}^{\text{state } f} \frac{RT}{p} dp = \int_{\text{state } i}^{\text{state } f} V_m(l) dP$$

- Initial state: no external pressure ($P = p = p^*$)
- Final state: $P = p^* + \Delta P$, $p = p$

$$\int_{p^*}^p \frac{RT}{p'} dp' = \int_{p^*}^{p^* + \Delta P} V_m(l) dP$$

Effect of ΔP on Vapor Pressure

$$\int_{p^*}^p \frac{RT}{p'} dp' = \int_{p^*}^{p^* + \Delta P} V_m(l) dP$$

Assuming $V_m(l)$ is constant,

$$RT \ln \left(\frac{p}{p^*} \right) = V_m(l) \Delta P$$

Therefore,

$$p = p^* e^{V_m(l) \Delta P / RT}$$

Brief Illustration 4B.2

For water, which has mass density 0.997 g cm^{-3} at 25°C and therefore molar volume $18.1 \text{ cm}^3 \text{ mol}^{-1}$, when the applied pressure is increased by 10 bar (*i.e.* $\Delta P = 1.0 \times 10^6 \text{ Pa}$), how much of vapor pressure will be increased?

Since $p/p^* = e^{V_{\text{m}}(\text{l})\Delta P/RT}$,

$$V_{\text{m}}(\text{l})\Delta P/RT$$

$$= (1.81 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}) \times (1.0 \times 10^6 \text{ Pa}) / (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) / (298 \text{ K})$$

$$= 0.0073$$

Hence, $p/p^* = e^{0.0073} = 1.0073$, that is, **0.73% increase**.